

SMART AS SHIT

Finding Data in Unexpected Places

What wastewater, broken traffic counters, and systems thinking can teach us about measuring the real world

SNOW
VISITORS
SMALL TOWN
BIG QUESTIONS

BIG BEAR LAKE
HIGHER INSIGHT
DEEPER TRUTH

SNOWMELT
INFILTRATION

GROUNDWATER
INFLOW

IN SPRING THE SEWER WAS NOT
COUNTING TOURISTS.
IT WAS COUNTING THE THAW.

In July it was counting people.

MORE WINTER WATER
IN THE SEWER

SAME PIPES.
A DIFFERENT STORY.

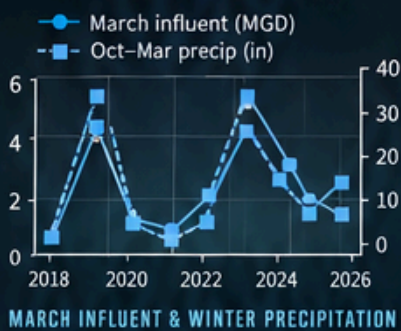
THE CONTROL

Peak-tourism July influent
stays nearly flat:
73–1.96 MGD (2018–2026).



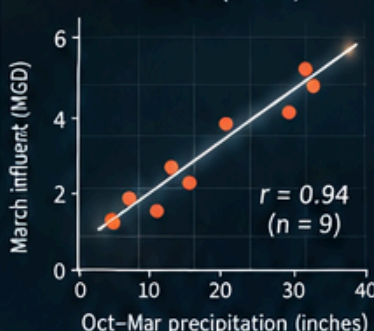
THE WEATHER YEAR

March influent tracks
winter water, not
visitor counts.



THE SIGNAL

Oct-Mar precipitation vs
March influent:
 $r = 0.94$ ($n = 9$).



THE MONSTER YEAR

2019: 4.48 MGD after
33.2 in winter precipitation
2023: 5.13 MGD after
31.3 in winter precipitation
and 83 in snowfall.

2019
4.48 MGD
33.2 in precip

2023
5.13 MGD
31.3 in precip
83 in snow



THE SYSTEM

Member shares stayed roughly
stable in March spike years:
53–62% Big Bear Lake /
35–44% CSD / 3–5% Fawnskin.



THE MODEL

Observed flow = sanitary
baseline + tourism load +
weather/I&I load.



THE TAKEAWAY

Wastewater can estimate transient
population — but only after separating
hydrologic infiltration and inflow
from human sanitary load.



Different Water.
Same Community.
Bigger Answer.

A MOUNTAIN. A SEWER. A TOURISM PROBLEM.
THE DATA WAS HIDING IN PLAIN SIGHT.

Research concept and analysis: Mason Perry